

SENTIMENT ANALYSIS — NEWS & SOCIAL MEDIA

SENTIMENT FOR IVV, SPY, AND QQQ

Overview

This project explores news and social media sentiment as a form of alternative data for financial analysis. The central question is whether the tone and mood of financial news articles and social media discussions around major ETFs — IVV, SPY, and QQQ — can be captured, quantified, and analyzed in a meaningful way.

Data was collected using the Alpha Vantage News Sentiment API, which aggregates real financial news articles and computes pre-built sentiment scores for each ticker. An additional layer of sentiment scoring was added using VADER — the Valence Aware Dictionary and Sentiment Reasoner — which is specifically designed for social and news media text.

The project produces six separate visualizations and a full correlation analysis of the sentiment features.

All code was written in Python and executed in Visual Studio Code.

DATA SOURCE

Alpha Vantage News Sentiment API Alpha Vantage provides a free API that pulls recent financial news articles from across the web and returns pre-computed sentiment scores for each article. For each article, the API returns:

- The article title and summary
- The publication source and date
- An overall sentiment score and label for the article
- A ticker-specific sentiment score showing how positive or negative the article is specifically about the requested ticker
- A ticker relevance score showing how central the ticker is to the article

Data was collected for three tickers — IVV, SPY, and QQQ — with up to 50 articles per ticker, giving a combined dataset of up to 150 articles.

VADER Sentiment Scoring On top of the Alpha Vantage scores, VADER was applied independently to each article's title and summary. VADER produces a compound score between -1.0 and +1.0, where scores above 0.05 are labelled Bullish, scores below -0.05 are labelled Bearish, and everything in between is labelled Neutral.

A weighted combined VADER score was computed for each article as follows:

- Title weight: 40%
- Summary weight: 60%

This weighting reflects the fact that article summaries tend to carry more substantive information than headlines alone.

FEATURES IN THE DATASET

Feature	Description
ticker	ETF ticker symbol
title	Article headline
summary	Article summary
source	News publication source
published	Article publication timestamp
overall_sentiment	Alpha Vantage overall sentiment score
overall_label	Alpha Vantage sentiment label
ticker_sentiment	Alpha Vantage ticker-specific score
ticker_relevance	How relevant the article is to the ticker
vader_title	VADER score for the title
vader_summary	VADER score for the summary
vader_combined	Weighted VADER score
vader_label	Bullish, Bearish, or Neutral

Methodology

Step 1 — Data Collection The Alpha Vantage API was called for each of the three tickers. For each article returned, all available sentiment fields were extracted and stored in a structured pandas DataFrame.

Step 2 — VADER Scoring VADER was applied to the title and summary of each article independently. A weighted combined score was then computed and used to assign a final sentiment label to each article.

Step 3 — Exploratory Data Analysis Six separate visualizations were produced to explore the distribution, trends, and relationships within the sentiment data. Each chart was saved as an individual PNG file.

Step 4 — Correlation Analysis A correlation matrix was computed across the four key numerical sentiment features to examine how the different scoring methods relate to each other.

RESULTS

Sentiment Distribution

Across all three tickers, the majority of articles were classified as Bullish by the VADER model. This is consistent with the general tendency of financial news to lean positive — negative articles tend to be more sensational but less frequent than neutral or positive coverage during stable market periods.

IVV showed the most balanced sentiment distribution of the three tickers, which makes sense given its nature as a broad market index fund. SPY and QQQ showed slightly more extreme sentiment distributions, possibly because they attract more active trading commentary.

Sentiment Over Time

The daily average VADER sentiment score fluctuated across the data collection period. There were visible periods where sentiment dipped into negative territory across all three tickers simultaneously — likely corresponding to broader market uncertainty events. These synchronized dips are interesting because they suggest that news sentiment around these ETFs moves together, reflecting their correlation as broad market instruments.

Alpha Vantage vs VADER Comparison

The scatter plot comparing Alpha Vantage sentiment scores to VADER scores showed a generally positive relationship — articles that Alpha Vantage scored as positive also tended to receive positive VADER scores. However there was meaningful scatter around the trend, which shows that the two methods do not always agree. This is expected because Alpha Vantage uses its own proprietary model trained specifically on financial text, while VADER is a general-purpose sentiment tool.

Relevance Score Distribution

The ticker relevance scores were heavily skewed toward higher values, meaning most articles returned by the API were genuinely and directly about the requested tickers rather than just mentioning them in passing. This is a good sign for data quality — the API is returning articles that are actually relevant to the investment question at hand.

VADER Score Distribution

The distribution of VADER combined scores was roughly bell-shaped and centered slightly above zero, confirming the overall positive lean of financial news sentiment. The mean score sat around 0.10 to 0.15 across the full dataset, indicating a mild but consistent bullish bias in the news coverage of these three ETFs.

Correlation Analysis

Feature Pair

Overall Sentiment vs Ticker Sentiment
Overall Sentiment vs VADER Combined
Moderate Positive
Ticker Sentiment vs VADER Combined
Ticker Relevance vs VADER Combined

Correlation

Moderate Positive
Weak to

Weak Positive
Near Zero

The correlation between the Alpha Vantage overall sentiment and ticker-specific sentiment was the strongest, which makes sense — articles that are generally positive tend to also be positive about the specific ticker they are discussing. The correlation between Alpha Vantage scores and VADER scores was weaker, confirming that the two methods capture different aspects of sentiment. Ticker relevance showed almost no correlation with sentiment score, meaning how relevant an article is to a ticker tells you nothing about whether the article is positive or negative.

VISUALIZATIONS

Plot 1 — Sentiment Distribution by Ticker

Bar chart showing the count of Bullish, Neutral, and Bearish articles for each of the three tickers side by side. Bullish dominated across all three.

Plot 2 — Daily Average Sentiment Over Time

Line chart showing how the average VADER sentiment score moved over the data collection period for each ticker. Synchronized dips across all three tickers were visible at certain points.

Plot 3 — Alpha Vantage vs VADER Scatter

Scatter plot comparing the two sentiment scoring methods article by article, color coded by ticker. A general positive trend was visible with meaningful scatter around it.

Plot 4 — Relevance Score Distribution

Histogram showing the distribution of ticker relevance scores across all articles. The distribution was skewed toward higher values confirming good data quality.

Plot 5 — VADER Score Distribution

Histogram showing the distribution of VADER combined scores across the full dataset. The distribution was roughly bell-shaped and centered slightly above zero.

Plot 6 — Correlation Heatmap

Heatmap showing pairwise correlations between the four key numerical sentiment features. The strongest correlation was between overall and ticker-specific Alpha Vantage scores.

FINDINGS

1. Financial news about major ETFs leans bullish:

The majority of articles across all three tickers were classified as positive. This is an important bias to be aware of when using news sentiment as a trading signal — raw sentiment scores may systematically overstate bullish conditions.

2. Alpha Vantage and VADER capture different things

The two scoring methods agreed directionally most of the time but differed on magnitude. Using both together gives a more complete picture than relying on either alone.

3. Sentiment across IVV, SPY, and QQQ moves together

The synchronized sentiment patterns across all three tickers confirm that news sentiment for broad market ETFs is driven by macro factors rather than fund-specific news. A piece of news that makes the market feel generally positive tends to lift sentiment for all three at once.

4. Relevance scores were consistently high

The Alpha Vantage API returned articles that were genuinely relevant to the requested tickers. This means the dataset is clean and the sentiment scores reflect actual coverage of these funds rather than tangential mentions.

5. VADER is a useful but imperfect tool for financial text

VADER was originally built for social media text and performs reasonably well on financial headlines and summaries. However it sometimes misses financial-specific language — for example it may score a phrase like "the fund saw heavy selling" as neutral rather than negative because it focuses on general word polarity rather than financial context.

RECOMMENDATIONS

1. Use sentiment as a complementary signal not a standalone one:

News sentiment data is most valuable when combined with price-based signals like those explored in the Financial Modelling notebook. Sentiment alone is noisy and subject to systematic biases. But combined with technical indicators, it can add meaningful context — especially around major market events.

2. Account for the bullish bias

Any model that incorporates raw sentiment scores from financial news should normalize for the systematic positive lean. One approach is to use sentiment relative to its own recent average rather than the absolute score.

3. Consider upgrading to a financial-specific NLP model

VADER is a good starting point but a model trained specifically on financial text — such as FinBERT — would produce more accurate sentiment classifications for this use case. This is a natural next step for extending this analysis.

4. Monitor sentiment velocity not just level

Rather than just looking at whether sentiment is positive or negative, tracking how fast sentiment is changing can be a more powerful signal. A rapid shift from bullish to bearish — even if the absolute level is still positive — may be an early warning of a market move.

5. Expand the ticker universe

This analysis covered three U.S. equity ETFs. Extending the same pipeline to emerging market ETFs like ECH and EWZ — the focus of the Financial Modelling notebook — would allow for a direct comparison of how sentiment dynamics differ between developed and emerging markets.

LIBRARIES USED

requests — API data collection
pandas — data structuring and manipulation
vaderSentiment — VADER sentiment scoring
matplotlib — individual plot generation
seaborn — correlation heatmap
numpy — numerical operations